國立陽明交通大學 National Yang Ming Chiao Tung University

2026 Spring Semester 隨機過程 Stochastic Processes course syllabus

Course name: (Chinese) 隨機過程 (English)Stochastic Processes				Dept. of course offering:	應數碩
				Course code:	SCMA30050
				Class Time/Room:	W34R8-SA213[GF]
Lecturers: 千野由喜					
credit:	3.00	type:	選修	grade:	*
Prerequisite: Lebesgue Integral, Measure Theory, Probability Theory, and Advanced Probability Theory are required.					
Course descriptions and objectives: Stochastic processes will broadly appear in a part of science, finance and social studies. The aim of this course is to understand stochastic process from the theoretical point of view and be familiar with how to handle it at the basic level. The course will be considered as an application of Probability Theory and Advanced Probability Theory. Thus we assume that students have been already familiar with measure-theory based probability theory.					
Textbooks: (Please indicate the book’ s name, author, publisher and the publication date):	We will have lecture note for each section of the course. To more knowledge or details, we will show some references below. -[Basic] R. Bass, (2012). Stochastic Processes, Cambridge University Press. -[Basic] R. Durrett, (2012). Essentials of Stochastic Processes, Springer. -[in Japanese] Y. Higuchi, and M. Nishio, (2006). Kakuritsukatei Nyumon, Baihu-kan. -[in Japanese] K. Ito, (2007). Kakuritsuron no Kiso [New edition], Iwanami-Shoten. -[Mathematical Finance] I. Karatzas, and S. E. Shreve, (2014). Brownian Motion and Stochastic Calculus, Springer.				

	-[Basic] G. F. Lawler, (2018). Introduction to Stochastic Processes, Chapman and Hall/CRC. -[Advanced] T. M. Liggett, (1985). Interacting Particle Systems, Springer. -[Basic] D. W. Stroock, (2005). An Introduction to Markov Processes, Springer. -[Random walk theory] W. Woess, (2000). Random Walks on Infinite Graphs and Groups, Cambridge University Press.

Course Contents		Hours				memo
Topics	Outlines	lectures	Demonstration	Practice	Others	

Description of Course Details:

1.Homework and Assignments, Exams and Quizzes, Evaluation and Grading Policy:

We will have report(s) and exam for evaluation.

2.Pedagogy and other supplementary information (websites, TAs, handouts and/or databases):

Office hours	Time slot	Location	Contact information
	TBA	SA239	Make an appointment by email: y.chino@math.nctu.edu.tw

Syllabus

Week	Date	Course Progress, Contents, Topics	Teaching Hours
1	2026-02-25(三),2026-02-26(四)	Introduction to Stochastic Processes & Review of Probability Theory	
2	2026-03-04(三),2026-03-05(四)	Review of Probability Theory	
3	2026-03-11(三),2026-03-12(四)	Discrete Time Stochastic Processes	
4	2026-03-18(三),2026-03-19(四)	Martingales: Random Walks	
5	2026-03-25(三),2026-03-26(四)	Martingales: Convergence Theorem	
6	2026-04-01(三	Markov Processes	

	),2026-04-02(四)		
7	2026-04-08(三),2026-04-09(四)	Markov Chains: Recurrence and Transience	
8	2026-04-15(三),2026-04-16(四)	Markov Chains: Irreducibility and Invariant distribution	
9	2026-04-22(三),2026-04-23(四)	Continuous Stochastic Processes	
10	2026-04-29(三),2026-04-30(四)	Continuous Stochastic Processes	
11	2026-05-06(三),2026-05-07(四)	Brownian Motion	
12	2026-05-13(三),2026-05-14(四)	Brownian Motion	
13	2026-05-20(三),2026-05-21(四)	Brownian Motion	
14	2026-05-27(三),2026-05-28(四)	Examination	
15	2026-06-03(三),2026-06-04(四)	Backup	
16	2026-06-10(三),2026-06-11(四)	Backup	

memo:

1.To protect intellectual property right, please do not illegally photocopy textbooks.

2.The others column of course details includes outside campus visit and seminar.

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